Meta-Learning-Based Semi-Supervised Change Detection in Remote Sensing Images

Yi Tang, Liyi Zhang, Wuxia Zhang, *Associate Member, IEEE*, and Zuo Jiang

***Abstract*— Semi-supervised change detection methods with consistency regularization, which overcome the lack of labeled samples by using unlabeled samples and enforcing consistent predictions under weak perturbations. However, current consistency regularization methods lack randomness in their perturbation settings and treat all samples uniformly, limiting the model's ability to leverage sample diversity to improve generalization. In contrast, meta-learning methods shift focus from individual samples to learning patterns across similar tasks, thereby enhancing information efficiency and model generalization. Inspired by these principles, we propose a meta-learning-based semi-supervised change detection (MLSCD) method for remote sensing images, which aims to explore and leverage meta-learning methods to enhance the generalization capabilities of consistency regularization-based semi-supervised change detection. First, we set the degree of weak perturbation and the combination of different types of perturbations as random parameters to generate diverse and randomized weak perturbations. Second, we redefine consistency regularization-based semi-supervised change detection from a meta-learning perspective, which learns patterns from diverse perturbation tasks to improve sample utilization efficiency, thereby enhancing the model's generalization capability. Third, we balance accuracy and efficiency by using AdamW for cross-task updates in the outer loop and stochastic gradient descent (SGD) for single-task optimization in the inner loop, which experimental results demonstrate is an ideal method for applying meta-learning to remote sensing change detection. Finally, the superiority of the proposed method is validated on two datasets. The extensive experimental results demonstrate the superior performance of the proposed method compared to state-of-the-art methods.**

***Index Terms*— Change detection, meta-learning, semi-supervised learning, task diversity.**

I. Introduction

CHANGE detection method [1], [2], [3], [4] in remote sensing images is crucial in environmental monitoring, prediction of natural disasters, urban development, and agricultural management. Change detection methods involve identifying and analyzing variations in data at different times. The effectiveness of current deep-supervised change detection methods is largely dependent on extensive labeled samples [5], [6], [7]. Insufficient labeled samples severely degrade change detection performance in remote sensing imagery, as the model struggles to capture data diversity, reducing generalization, and detection accuracy. In emergency natural disasters, acquiring labeled samples for bi-temporal remote sensing images is particularly challenging. Manual comparisons are required to obtain change labels, which requires ample annotated training samples a difficult task. Semi-supervised change detection methods have gained popularity for reducing the model's dependence on labeled samples, which effectively utilize large amounts of unlabeled data to compensate for the limited labeled samples.

Current semi-supervised change detection methods generate pseudo-labels of unlabeled samples using self-training, generative adversarial networks (GANs), variational autoencoders (VAEs), graph-based learning, and entropy minimization to reduce dependency on labeled samples [8], [9]. However, these pseudo-label generation methods often encounter issues like label instability and error propagation. Semi-supervised change detection [10], [11], [12] with consistency regularization mitigates overfitting due to insufficiently labeled samples. The consistency regularization [13], [14] assumes that weak perturbations applied to the change feature map will not affect the change detection results. Consistency regularization methods [15], [16] reduce the dependency on labeled data and pseudo-labels by enforcing consistent model predictions on perturbed and unperturbed unlabeled data, effectively utilizing unlabeled samples for training. Remote sensing images are often affected by degradation, noise, and spectral variations during the imaging process, arising from atmospheric conditions, lighting, and sensor noise [17]. Learning from these unlabeled samples with perturbations not only reduces the model's reliance on labeled data but also enhances its robustness to real-world perturbations. However, current consistency regularization methods limit the diversity of perturbation learning, which is constrained by the use of fixed-intensity weak perturbations and the separate application of each perturbation type on unlabeled samples. Furthermore, treating all samples as a single task during training does not efficiently utilize the available data. Such a method overlooks the inherent randomness and diversity of real-world perturbations, which results in a model that is optimized only

Received 14 October 2024; revised 30 November 2024; accepted 2 January 2025. Date of publication 9 January 2025; date of current version 23 January 2025. This work was supported in part by the National Natural Science Foundation of China Funded Project under Grant 62462069, in part by the Scientific Research Fund Project of the Education Department of Yunnan Province under Grant 2024Y467, in part by the National Natural Science Foundation of China General Program under Grant 62471389, and in part by Shaanxi Provincial Key Research and Development Programme General Project under Grant 2024SF-YBXM-572. *(Corresponding author: Wuxia Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3527483

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for these specific perturbations and cannot be generalized to a wider range of conditions.

Inspired by meta-learning [18], [19], [20], [21], [22], [23], [24], samples are divided into multiple diverse tasks to enhance sample usage efficiency. Common patterns are learned from diverse tasks to improve the generalizability of the model. From a meta-learning perspective, semi-supervised change detection based on consistency regularization is redefined by treating the learning of patterns from perturbation-induced tasks as a meta-learning objective. We propose a meta-learning-based semi-supervised change detection (MLSCD) method for remote sensing images, which aims to enhance the robustness of semi-supervised change detection based on consistency regularization by leveraging meta-learning methods to partition all sample features into diverse learning tasks, thereby improving the efficiency of sample usage efficiency. Fixed predefined perturbations narrow the learning tasks, limiting the model to specific conditions and reducing its ability to generalize to new cases. To address these limitations of fixed perturbations, the meta-learning method generates diverse tasks by randomizing perturbation types and intensities. Through inner and outer loop training, the meta-learning [25] framework allows the model to learn from multiple tasks, identifying and extracting shared patterns in different perturbations, rather than being restricted to a single fixed task. The following is a detailed process for building meta-learning tasks.

The task generation process involves selecting different types of perturbation, such as noise, scaling, rotation, and translation, to modify the feature differences of unlabeled sample pairs. The intensity of the perturbation is randomized to increase the diversity of tasks, and various types of perturbation are randomly combined to create more complex tasks. In addition, these perturbation types and intensities collectively form a perturbation space from which various tasks are generated by random sampling. This article ensures that the combination of multiple weak perturbations remains weak by limiting the intensity of each perturbation and using a weighted combination to prevent them from summing into a strong perturbation. As illustrated in Fig. 1, incorporating randomized combination perturbations significantly enhances the randomness and diversity of the perturbations, more closely mirroring the perturbations encountered in real-world scenarios. Each task contains a perturbed feature difference map of unlabeled pairs, an unperturbed feature difference map of unlabeled pairs, and a feature difference map of labeled pairs. Perturbations are applied to the unlabeled sample pairs and combined with the labeled pairs to create diverse learning tasks, with the randomness of the perturbations ensuring the generation of diverse tasks. For diverse learning tasks, the next step is to design the inner and outer loops of meta-learning to make the meta-learning method more effective for change detection.

To balance the accuracy and efficiency of the model, the meta-learning method employs stochastic gradient descent (SGD) [26], [27], [28], [29] in the inner loop and adaptive moment estimation weight decline (AdamW) [30] in the outer loop. In the inner loop of SGD, parameter updates rely solely on the gradient of the current sample, enabling rapid learning of meta-knowledge for each task. This SGD eliminates the need for complex momentum and second-order moment estimates, thereby reducing the computational overhead per update and achieving faster optimization. The outer loop utilizes AdamW to refine meta-knowledge acquired from the inner loop, optimizing model parameters such as adaptive learning rates and weight decay for improved generalization across various tasks. In meta-learning, the inner loop acquires meta-knowledge by identifying patterns across different task sets, while the outer loop refines this knowledge by continually validating it with labeled tasks.

In summary, our primary contributions are: 1) we set the degree and combination of different types of weak perturbations as random parameters to generate diverse and randomized perturbations, which perturb the feature differences of unlabeled sample pairs to create diverse learning samples; 2) we redefine consistency regularization-based semi-supervised change detection from a meta-learning perspective, which divides samples into diverse learning tasks to improve sample utilization efficiency, extracts patterns from these tasks significantly and enhances the robustness and generalization ability of the change detection model; and 3) we found that using AdamW for cross-task updates in the outer loop and SGD for single-task optimization in the inner loop effectively balances accuracy and efficiency in remote sensing change detection, as demonstrated by the experimental results.

II. Related Work

In recent years, deep learning has significantly advanced remote sensing [17], particularly in spectral data processing, multisource data fusion, hyperspectral image classification, and change detection. Methods such as GANs, Transformers, and GNNs enable deep exploration of complex data patterns, leading to efficient feature extraction and data fusion. For example, spectralGPT [31] utilizes a 3-D generative pretrained transformer (GPT) to enhance the integration of spatial and spectral features for remote sensing applications. HighDAN [32] improves the generalization of the model across tasks by leveraging high-to-low resolution fusion and adversarial learning. CasFormer [33] enhances spatial and spectral resolution by fusing RGB and hyperspectral data using cascaded attention mechanisms. MorphFormer [34] strengthens feature extraction by incorporating learnable spatial and spectral morphological convolutions into the self-attention mechanism. Additionally, combining CNNs with RNNs improves feature extraction by capturing spatial features through CNNs and temporal dependencies through RNNs, especially for complex tasks [35]. However, when labeled data are limited, maintaining generalization becomes a challenge. Domain adaptation improves generalization by aligning feature representations between the source and target domains [36], while the dual teacher framework takes advantage of semisupervised learning to effectively use unlabeled data [37]. Rotation-Invariant Attention Networks enhance robustness to data variability, such as image orientation [38], and multitask learning increases efficiency on small datasets by sharing feature representations across tasks [39]. These deep learning

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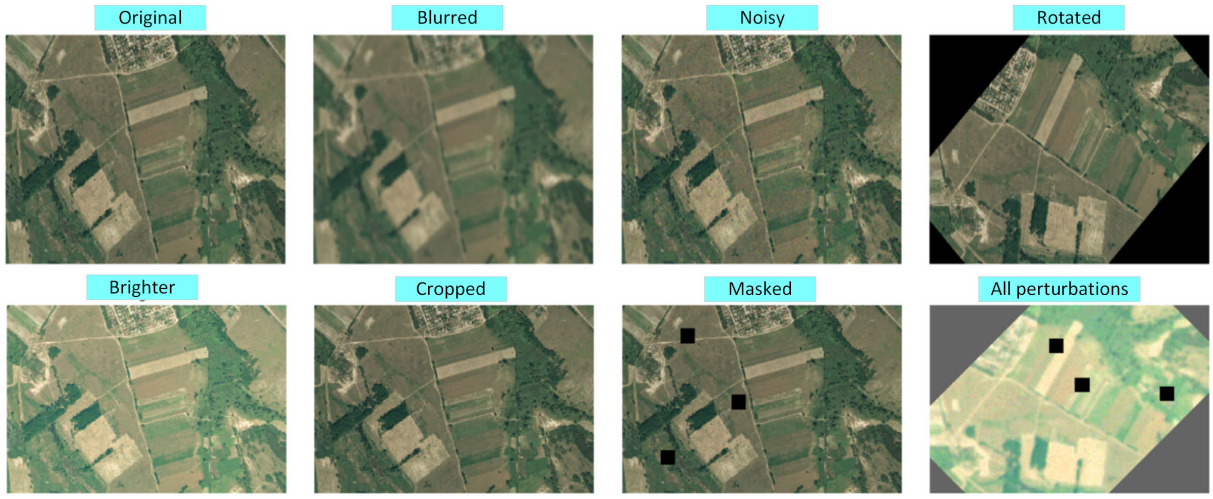


Fig. 1. Visualizations of all types of weak perturbations and their combinations.

advancements form a solid foundation for remote sensing image change detection and are crucial for ongoing research.

A. Change Detection

With the advancement of deep learning methods, change detection has garnered increasing attention as a core focus in remote sensing applications. Supervised change detection methods typically require comparing prechange and postchange images to obtain a change map. The main existing approaches are: early fusion, which processes fused prechange and postchange images along the spectral dimension [40], [41], [42]. Siamese concatenation, which concatenates hidden features from prechange and postchange images [43], [44]. The Siamese difference, which processes the absolute difference of hidden features [45], [46] with the Siamese difference showing better results. However, supervised methods heavily rely on large labeled samples and cannot leverage abundant unlabeled remote sensing samples. To address this, researchers have explored semi-supervised learning for change detection. The goal is to effectively use information from unlabeled remote sensing samples to improve performance while reducing reliance on large labeled samples.

In the field of semi-supervised change detection in remote sensing images, Ghosh et al. [47] used the membership values of its K nearest neighbors to generate soft class labels. Yuan et al. [48] addressed the problem of insufficient samples by using semi-supervised Laplacian regularization metric learning. With the wide application of GANs in image-processing tasks, GDCN [49] used GANs to generate fake samples using random noise for the training of change detection models. Although this method reduces the model's dependence on labeled samples, it does not make use of unlabeled samples. To take advantage of unlabeled remote sensing images, Peng et al. [50] proposed SemiCD, which uses GANs to improve the quality of pseudo-labels generated by the model and finally improves model performance. Wang et al. [12] designed a contrastive loss function based on changed areas and selected reliable pseudo-labels to improve the performance of semi-supervised change detection models. In addition, Han et al. [13] proposed a coarse-to-fine semi-supervised change detection method based on consistency regularization (C2F-SemiCD).

In recent works, Bandara and Patel [10] introduced perturbations into the middle layer of the change detection network and trained the consistency loss of predicted outputs with perturbations to improve the robustness and feature extraction ability of the detection model. Yang et al. [11] proposed a method to improve the consistency between weakly and strongly labeled samples to better utilize unlabeled samples in semantic segmentation and change detection tasks. In addition, DiffMatch [51] is a semi-supervised change detection method guided by a vision–language model. However, traditional consistency regularization methods in semi-supervised detection rely on fixed and predetermined perturbation parameters, failing to account for the randomness and diversity of real-world perturbations, thus limiting task diversity. This study explores the generation of diverse tasks through constructing perturbations with stochasticity and variety. Utilizing meta-learning for perturbation parameters and task diversity has proven effective. Below, we discuss related research in meta-learning for parameter learning.

B. Meta-Learning

Meta-learning methods of parameter learning [18], [23], [52], [53] aim to "learn to initialize," allowing the base model to rapidly learn task-specific knowledge in the inner loop stage. In the outer loop stage, the meta-model acquires common knowledge across tasks, acknowledging similarities among various tasks. This dual-loop design enables the model to quickly adapt to new tasks by learning both task-specific and common knowledge. Finn et al. [21] proposed the model-agnostic meta-learning (MAML) method, which trains a model on multiple tasks to optimize initial parameters and enhance generalization ability. Antoniou et al. [54] proposed MAML++, which incorporates multistep Loss optimization for more fine-grained parameter adjustment. Meta-learning methods are widely used and are highly effective. Naman and Deepshikha [55] used MAML algorithms to meta-learning initial parameters from samples in multiple languages,

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enabling fast adaptation to unseen target languages. In addition, Lee et al. [56] proposed a modeling method based on deep learning and MAML, which solves the adaptability problem in action unit (AU) detection. Furthermore, the FCNN framework proposed by Zhu et al. [57] aims to address the problem of personalized blood glucose prediction in T1D. Lastly, Chen et al. [58] proposed a simple meta-learning procedure on their pretrained model for evaluating the overall classification. However, there is currently scarce research on the use of meta-learning for remote sensing image change detection.

III. Method

This article employs a meta-learning framework, as illustrated in Fig. 2. The framework begins by partitioning each dataset into support and query sets. The support set comprises partially labeled and entirely unlabeled samples, which serve to enrich the model learning capability through unlabeled samples. Meanwhile, the query set consists of the remaining labeled samples, essential for evaluating the model's performance on novel tasks. During the meta-training phase, the model undergoes training on each task in the support set, thereby acquiring meta-parameters conducive to handling diverse perturbations and fortifying its robustness. Concurrently, the outer loop leverages the query set to refine perturbation parameters, further fine-tuning the model. Subsequently, in the Meta-Testing phase, the optimized model parameters are put to the test on fresh evaluation tasks to gauge its generalization prowess. Through the coordinated interplay of these components, the framework aims to enhance change detection efficacy in remote sensing imagery under conditions of limited labeled samples. Below, a detailed overview of these three modules is provided.

A. Splitting Dataset

In the meta-learning framework, the dataset $\mathcal{D}$ is divided into a labeled set $\mathcal{D}_l$ and an unlabeled set $\mathcal{D}_{ul}$, with $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_{ul}\}$. The labeled set $\mathcal{D}_l$ is randomly split into two subsets: a query set $\mathcal{D}_q$, which comprises $a\%$ of $\mathcal{D}_l$, and a support set $\mathcal{D}_s$, which includes the remaining $1 - a\%$ of $\mathcal{D}_l$ combined with the entire unlabeled set $\mathcal{D}_{ul}$. The query set is reserved for evaluation, while the support set is used for training. The partitioning process is expressed as

$$\begin{aligned} \mathcal{D}_q &= f_s(\mathcal{D}_l, a\%) \\ \mathcal{D}_s &= \{(1 - a\%)\mathcal{D}_l, \mathcal{D}_{ul}\} \\ \mathcal{D} &= \{\mathcal{D}_q, \mathcal{D}_s\} \end{aligned} \tag{1}$$

where $f_s(\mathcal{D}_l, a\%)$ represents the random selection of $a\%$ of labeled samples for $\mathcal{D}_q$. This data partitioning approach divides the support set and query set well, preparing for meta-training. The partitioning also ensures that distinct data samples are used for training and validation, thereby improving the model's generalization ability on new tasks.

B. Meta-Training

The meta-learning method achieves a balance between model accuracy and efficiency by using SGD in the inner loop while adopting adaptive moment estimation with weight decay (AdamW) in the outer loop. The inner loop rapidly learns task-specific patterns from the support set to update the model parameters, while the outer loop evaluates the generalization capability of the updated parameters on the query set and makes necessary adjustments. Detailed algorithmic steps are provided in Algorithm 1. The SGD method quickly adapts to each task from the support set, reducing overfitting by relying solely on the gradient calculation for the current sample, without involving complex momentum or second-order moment estimates. This minimizes the computational load per update, expediting optimization. Additionally, SGD's simplicity helps prevent task-specific overfitting, allowing rapid adaptation to diverse task patterns and improving generalization on new tasks. The algorithm detailed in Algorithm 2 involves acquiring distinct tasks perturbed from the support set, followed by gradient computation and parameter updates to effectively optimize model parameters. The outer loop uses the AdamW optimizer to adjust the learning rate based on past gradients, improving the adaptability to diverse tasks. Additionally, AdamW incorporates weight decay to reduce overfitting and enhance model generalization. The detailed update procedure is provided in Algorithm 3. In the outer loop, AdamW efficiently manages model complexity and updates the parameters obtained from the inner loop using feedback from the query set, enabling effective adaptation across diverse tasks. Below, detailed explanations of the inner and outer loop algorithms are provided.

1) Inner Loop Optimization: ① Feature difference map extraction. Diverse tasks selected from the support set provide the model with fundamental information for learning. By capturing features and patterns from diverse tasks, the model becomes capable of making accurate predictions on the subsequent query set. Feature difference map extraction begins by initializing the model parameters θ. For each task t in the support set, a subset of labeled samples $(I_{lA,i}, I_{lB,i}, y_i)$ is randomly selected from the supervised dataset D_l, and a subset of unlabeled samples $(I_{\text{ulA},i}, I_{\text{ulB},i})$ is randomly selected from the unsupervised dataset D_{ul}. The hidden feature representations F_A and F_B are computed for the labeled image pair $(I_{lA,i}, I_{lB,i})$, while F_{uA} and F_{uB} are computed for the unlabeled image pair $(I_{\text{ulA},i}, I_{\text{ulB},i})$. Feature mappings transform the visual data from each image into a representative feature space, as mathematically formulated

$$F_A = f_e(I_A), \quad F_B = f_e(I_B) \tag{2}$$

$$F_{uA} = f_e(I_{uA}), \quad F_{uB} = f_e(I_{uB}) \tag{3}$$

where F_A and F_B represent the feature representations of the labeled sample pair, and F_{uA} and F_{uB} represent the feature representations of the unlabeled sample pair. The hidden feature difference maps are calculated as follows:

$$F_d = f_{\text{PPM}}(|F_A - F_B|), \quad F_{ud} = f_{\text{PPM}}(|F_{uA} - F_{uB}|) \tag{4}$$

where the feature difference map F_d is obtained from the labeled representations F_A and F_B, and F_{ud} is derived from the unlabeled representations F_{uA} and F_{uB}.

② During the supervised phase. The parameters of the change detection network are optimized by extracting labeled

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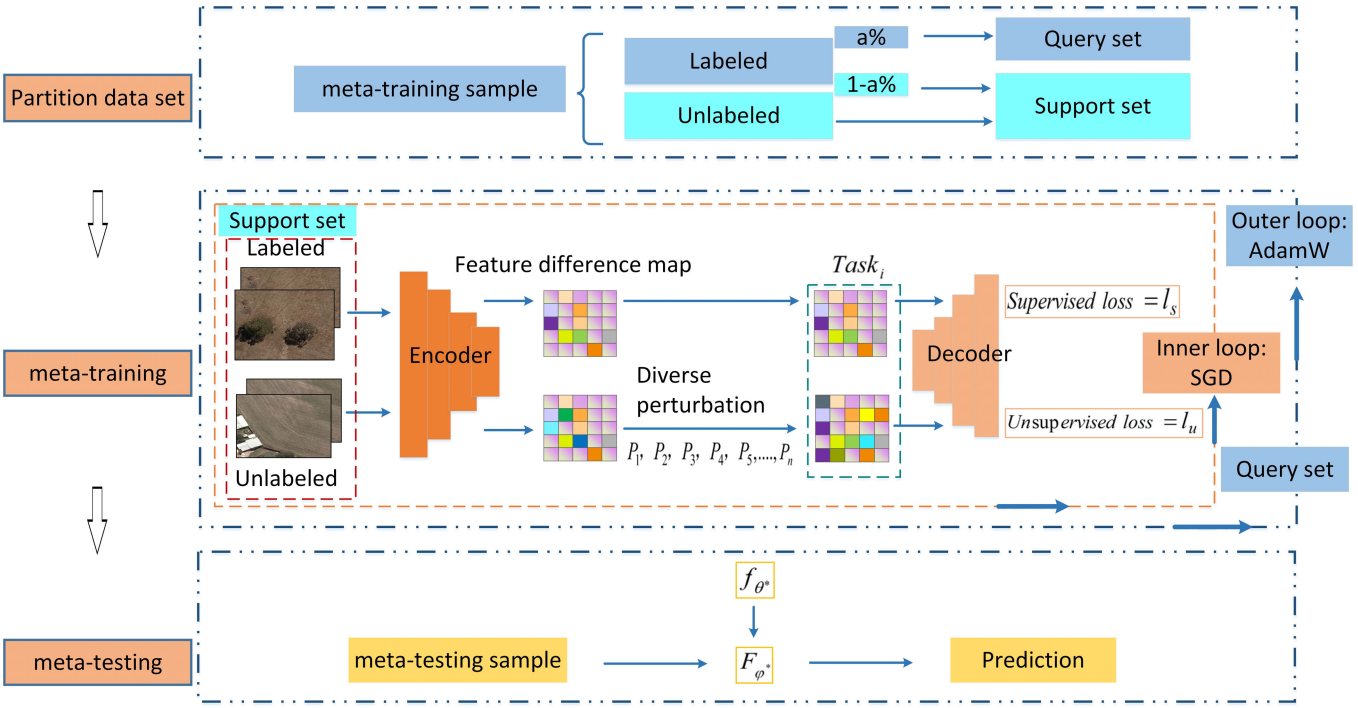


Fig. 2. Flowchart of meta-learning-based semi-supervised change detection in remote sensing images.

samples from the support set according to (1). Labeled samples in the meta-learning inner loop are expressed as

$$(1-a\%)\mathcal{D}_l = \left\{ \left(I_{A,i}^l, I_{B,i}^l, y_i^l \right) \mid i = 1, \ldots, N_l \right\} \tag{5}$$

where $(I_{A,i}^l, I_{B,i}^l)$ denotes the ith bi-temporal image pair and y_i^l is the corresponding ground-truth change mask, with N_l representing the dataset size. The hidden feature difference F_{di} is derived from (4), and the change probability map $\hat{y}_i$ is estimated from F_{di} using the labeled sample pairs. This process in the decoder can be mathematically expressed as

$$\hat{y}_i = f_d(F_{di}) \tag{6}$$

where f_d is the decoder of the change detection network. The supervised loss is computed using the cross-entropy loss function, given by

$$L_{\text{sup}} = \text{CE}\left(\hat{y}_i, y_i\right) \tag{7}$$

where CE denotes the cross-entropy loss function between $\hat{y}_i$ and y_i.

③ During the unsupervised phase. Parameters of the change detection network are optimized by extracting unlabeled samples from the support set according to (1). Unlabeled samples in the meta-learning inner loop are expressed as

$$a\%\mathcal{D}_{ul} = \left\{ \left(I_{A,i}^{ul}, I_{B,i}^{ul} \right) \mid i = 1, \ldots, N_{ul} \right\} \tag{8}$$

where $(I_{A,i}^{ul}, I_{B,i}^{ul})$ denotes the ith bi-temporal image pair, with N_{ul} representing the dataset size. The hidden feature difference F_{di}^{ul} is derived from (4). For each pair unlabeled sample $(I_{A,i}^{ul}, I_{B,i}^{ul})$, the feature difference map F_{di}^{ul} is subjected to stochastic and diverse perturbations. Perturbations are applied to feature maps instead of images to enhance robustness by capturing high-level semantics and reducing irrelevant changes.

Real-world remote sensing images often suffer from noise, occlusions, and information loss due to atmospheric, lighting, and sensor factors. To simulate these perturbations, five types of perturbations are selected, as shown in Fig. 3. The construction of the perturbation parameter space begins with the deliberate selection of perturbation types, such as noise, geometric distortions, and intensity adjustments, which are chosen for their relevance to real-world conditions. The ranges for these perturbations are defined based on empirical data and domain knowledge to ensure that they reflect realistic scenarios. Each perturbation type p_i ($i = 1$ to 5) is mathematically defined as

$$p_{it} = f_r(m_i, n_i) \tag{9}$$

where f_r randomly selects values within the specified range (m_i, n_i). This randomization process is critical for introducing variability in the perturbation space. The selected perturbations establish the initial parameter space P_1

$$P_1 = (p_1, \ldots, p_5). \tag{10}$$

To increase the diversity and complexity of the perturbation space, these perturbations are combined randomly

$$p_{ti} = C_5^{p_i} \tag{11}$$

where C represents the random combination of perturbation types. This combination evolves P_1 into a more complex perturbation space P_2

$$P_2 = \left(p_1, \ldots, p_5, p_{t1}, \ldots, p_{tn}\right) \tag{12}$$

enhancing its dimensionality and diversity, which ensures that the model can generalize to a wide range of real-world scenarios.

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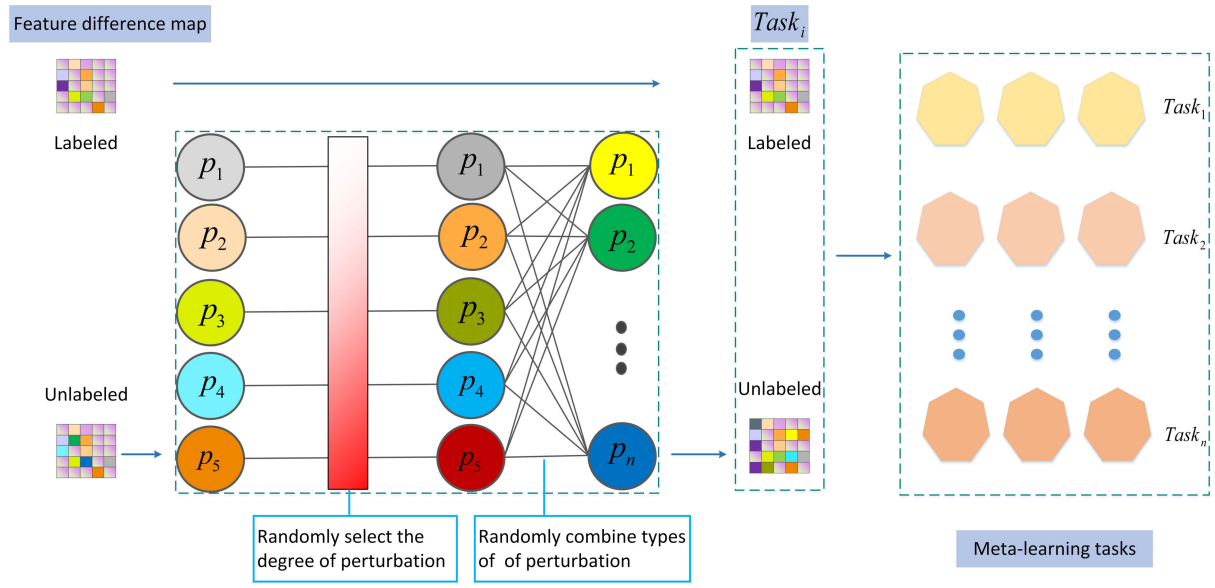


Fig. 3. Flowchart of stochastic and diverse perturbations.

Algorithm 1 Meta-Learning

Input: Supervised dataset D_l, Unsupervised dataset D_{ul}, Percentage of labeled data $a\%$, Inner loop optimizer (SGD), Outer loop optimizer (AdamW)

Output: Optimized model parameters θ^*, φ^*

Initialize model parameters θ
Partition data set into meta-training samples and a query set
Divide meta-training samples into labeled $(a\%)$ and unlabeled $(1-a\%)$
for each task T_{ask} in the meta-training set **do**
  Randomly select a subset of labeled samples $(I_{lA,i}, I_{lB,i}, y_i)$ from D_l
  Randomly select a subset of unlabeled samples $(I_{ulA,i}, I_{ulB,i})$ from D_{ul}
  Encode labeled and unlabeled images to obtain feature maps
  Compute feature difference map for each input pair
  Apply random perturbation and combination to feature maps to generate perturbed versions $P_1, P_2, \ldots, P_n$
  Form task T_{ask_i} using one of the perturbed feature maps
  Decode task T_{ask_i} to produce outputs
  Calculate supervised loss l_s using labeled data
  Calculate unsupervised loss l_u using unlabeled data
  Update model parameters θ in inner loop using SGD to minimize l_s and l_u
end for
Update model parameters θ in outer loop using AdamW
Meta-Testing:
Use optimized parameters θ^* and φ^* for prediction on meta-testing samples
return Optimized model parameters θ^*, φ^*

Algorithm 2 Algorithm of the Inner Loop Optimizer

for For each iteration $t = 1, 2, \ldots, T$: **do**
  for For each sample $(x_i, y_i) \in D$: **do**
    Compute the gradient of the loss function: $g_t = \nabla_w L(w_{t-1}; x_i, y_i)$.
    Update the model parameters: $\theta_t = \theta_{t-1} - l_{ir_t} g_t$.
  end for
end for
Compute the task-specific initialization

Here, $\nabla_w L(w_{t-1}; x_i, y_i)$ represents the derivative of w with respect to the loss function, and θ_t is the model parameter

For each pair unlabeled $(I^{ul}_{A,i}, I^{ul}_{B,i})$, the feature difference map $F_{ud,i}$ is generated by (4). Perturbations in the diverse and stochastic space P_2 yield numerous perturbed feature difference maps

$$\tilde{F}_{ud,i,p} = f_p\left(F_{ud,i}, p\right), \quad \forall p \in P_2 \tag{13}$$

where $f_p(F_{ud,i}, p)$ applies perturbation p from P_2 to the feature difference map $F_{ud,i}$. From the perturbation space P_2, m perturbations are randomly selected and applied, producing m perturbed feature maps $\{\tilde{F}^{ul}_{d,1,p}, \tilde{F}^{ul}_{d,2,p}, \ldots, \tilde{F}^{ul}_{d,m,p}\}$. For each perturbation $\tilde{F}^{ul}_{d,i,p}$, the decoder $f_d(\cdot)$ predicts the change probability map $\hat{y}_{uli}$ as follows:

$$\hat{y}_{\text{uli}} = f_d\left(\tilde{F}^{ul}_{d,i,p}\right). \tag{14}$$

Simultaneously, the decoder $f_d(\cdot)$ predicts the probability map $\tilde{y}_{\text{uli},p}$ for the unperturbed feature difference map

$$\tilde{y}_{\text{uli},p} = f_d\left(\tilde{F}^{ul}_{d,i}\right). \tag{15}$$

According to the principle of consistency, minor perturbations should not affect the outcomes of change detection predictions. If the predicted results for $\hat{y}_{\text{uli}}$ and $\tilde{y}_{\text{uli},p}$ are identical, the unsupervised loss is represented as

$$L_{\text{unsup}} = \frac{1}{N_p} \sum_{p=1}^{N_p} d\left(\tilde{y}_{\text{uli},p}, \hat{y}_{\text{uli}}\right) \tag{16}$$

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Algorithm 3 Algorithm of the Outer Loop Optimizer

Input: Learning rate α, first moment decay rate β_1, second moment decay rate β_2, weight decay λ

Initialize time step $t = 0$, momentum variables $m_0 = 0$, $v_0 = 0$
while Parameters θ not converged **do**
  $t = t + 1$
  Compute gradient g_t
  Update biased first moment estimate: $m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$
  Update biased second moment estimate: $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$
  Correct bias of first moment: $\hat{m}_t = \frac{m_t}{1-\beta_1^t}$
  Correct bias of second moment: $\hat{v}_t = \frac{v_t}{1-\beta_2^t}$
  Update parameters: $\theta_t = \theta_{t-1} - \frac{l_{or}}{\sqrt{\hat{v}_t}+\epsilon} \cdot (\hat{m}_t + \lambda \cdot \theta_{t-1})$
end while
return Converged parameters θ

Here, g_t is the current gradient, m_t and v_t are the first and second moment estimates, and $\hat{m}_t$ and $\hat{v}_t$ are their bias-corrected versions. t denotes the iteration number, θ_t are the model parameters, and l_{or} is the learning rate. β_1 and β_2 are exponential decay rates, λ is the weight decay coefficient, and ϵ prevents division by zero.

where $d(\cdot)$ measures the distance between $\hat{y}_{\text{uli}}$ and $\tilde{y}_{\text{uli},p}$.

Meta-learning tasks: consist of feature difference maps derived from both labeled and unlabeled sample pairs. The feature difference maps from unlabeled pairs are subjected to varying perturbations, creating a diverse set of meta-learning tasks. This diversity guides change detection models to learn common patterns across varied perturbation tasks, thus enhancing their generalization performance. The inner loop loss function is expressed as

$$L_{ir} = w_s L_s + w_u L_u \tag{17}$$

where L_s represents the supervised loss, L_u denotes the unsupervised loss, and the model parameters θ are updated in the inner loop to minimize L_{ir}.

2) In the Outer Loop: The query set evaluates the model's ability to generalize to new tasks. Performance on the query set indicates whether the model has successfully learned effective features from the support set. The process begins by taking the model parameters θ obtained from the inner-loop optimization as initial parameters. For each task t in the query set, a subset of samples $(I_{lA,i}, I_{lB,i}, y_i)$ is randomly selected from the supervised dataset D_l. The hidden feature representations F_A and F_B of the input image pair $(I_{lA,i}, I_{lB,i})$ are computed. Specifically, f_e processes the prechange image I_A and the postchange image I_B as inputs, generating their respective hidden feature representations F_A and F_B

$$F_A = f_e(I_A) \tag{18}$$

$$F_B = f_e(I_B). \tag{19}$$

The hidden feature difference is denoted as

$$F_d = f_{\text{PPM}}(|F_A - F_B|). \tag{20}$$

The change map $\hat{y}$ is predicted using the hidden feature difference F_d. The mathematical representation of this process in the feature difference module is given by

$$F_d = f_{\text{PPM}}(|F_A - F_B|). \tag{21}$$

The supervised loss is represented as

$$L_{\text{sup}} = \text{CE}\left(\hat{y}, y_i\right) \tag{22}$$

where it is computed using the cross-entropy loss function and added to the total loss L_t. After processing all tasks in the query set, the average loss L_a is calculated as

$$L_a = \frac{L_t}{N} \tag{23}$$

where N is the number of tasks.

For the model parameters θ obtained in the inner loop, the AdamW optimizer is employed to minimize L_a. Equations (10) and (16) are used in the outer loop to further validate and fine-tune the perturbed parameters using a query set. This approach prevents excessive complexity in the perturbed parameters, thereby yielding richer and more accurate meta-knowledge.

C. Meta-Testing

In the meta-testing phase, the optimized model parameters θ^*, developed from the meta-training process, are utilized. For each new task T_n, previously unencountered during training, task-specific parameters θ_{T_n} are initialized directly from θ^*. From the test dataset D_{test}, a selected subset of samples is given by

$$\left(I_{tA,i}, I_{tB,i}, y_{t,i}\right) \tag{24}$$

where it is processed for task T_n. Hidden feature representations F_{tA} and F_{tB} are computed, and the hidden feature difference F_{test_d} is calculated as

$$F_{\text{test}_d} = f_{\text{PPM}}(|F_{tA} - F_{tB}|). \tag{25}$$

The change map $\hat{y}_t$ is predicted using this feature difference, and the test loss L_t is determined using the cross-entropy function. The model's performance on T_n is evaluated using metrics such as accuracy or intersection over union (IoU), assessing its capability to generalize and apply the learned meta-knowledge to new tasks. This evaluation, conducted in a variety of new tasks, provides a comprehensive insight into the model's effectiveness in meta-testing and its ability to generalize in diverse scenarios.

IV. Experiments

A. Datasets

This article used two publicly available remote sensing change detection datasets for experiments.

1) *LEVIR Dataset [59]:* This large-scale benchmark dataset is widely used to evaluate deep learning-based

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change detection algorithms. It consists of 10 192 high-resolution image pairs, each with a resolution of 256 × 256 pixels, and is divided into training and testing sets. The dataset is highly relevant to this study due to its comprehensive coverage of diverse urban environments, making it a representative choice for assessing change detection performance.

2) *WHU Dataset [60]:* This dataset contains 7434 high-resolution, multitemporal remote sensing image pairs, also at a resolution of 256 × 256 pixels. Similar to the LEVIR dataset, it is divided into training and testing sets. Its inclusion provides additional validation due to its focus on complex spatial changes, ensuring a robust evaluation of the model's ability to generalize across different types of remote sensing imagery.

Both datasets are selected for their relevance in covering diverse scenarios and ensuring the robustness of the change detection methods being validated. The datasets are processed into nonoverlapping 256 × 256 pixel blocks and divided into support and query sets. The notation $X \rightarrow Y$ indicates that the model is trained on dataset X's training set and tested on dataset Y testing set. In supervised learning experiments, models are trained with varying proportions (5%, 10%, 20%, and 40%) of labeled data, with the remainder unlabeled.

B. Experimental Setup

1) Parameter Setting: This study implements the proposed MLSCD method using the PyTorch framework on an RTX 2080 Ti GPU with 30 GB of memory. Training spans 80 epochs per dataset, with a learning rate of 0.001 and a batch size of 4. An enhanced AdamW optimizer with exponential weighting coefficients $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is employed. In the experiment, the parameter a represents the proportion of labeled samples in the query set. The parameter a is set to 5%, 10%, 20%, and 30% to evaluate the model's performance.

2) Evaluation Criteria:

1) *Recall (Rec.):* The ratio of correctly identified positive instances to the total actual positives, reflecting the model's sensitivity. The formula is written as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{26}$$

where TP denotes true positive and FN refers to false negative.

2) *IoU:* The ratio of the intersection to the union of the predicted results and ground truth, which measures the overlap between prediction and ground truth. A higher IoU indicates a more accurate prediction result. The formula is written as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{27}$$

where TP denotes true positive, FP means false positive, and FN refers to false negative.

3) *F1-score (F1):* The harmonic mean of precision and recall, providing a single metric that balances both concerns. The formula is written as follows:

$$\text{F1} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{28}$$

where recall and precision are as defined in (26) and (27).

4) *Kappa Coefficient (Kappa):* A statistical measure of interrater agreement or accuracy adjusted for chance agreement. The formula is written as follows:

$$\text{Kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}} \tag{29}$$

where the overall accuracy (OA) is calculated as

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{30}$$

and PRE is the probability of random agreement, given by

$$\text{PRE} = \frac{(\text{FP} + \text{TP}) \times (\text{FN} + \text{TP})}{(\text{TN} + \text{FP} + \text{TP} + \text{FN})^2} + \frac{(\text{TN} + \text{FP}) \times (\text{FN} + \text{TN})}{(\text{TN} + \text{FP} + \text{TP} + \text{FN})^2} \tag{31}$$

where TP denotes true positive, TN represents true negative, FP means false positive, and FN refers to false negative.

C. Experimental Results and Analysis

1) Comparison With State-of-the-Arts: This section evaluates the performance of the MLSCD method in various datasets, which is compared with the baseline SemiCD method using different optimizers and semi-supervised methods. Table I provides a detailed description of optimizer methods, while Table II presents a detailed overview of semi-supervised methods.

The optimization advantages of the MLSCD method are demonstrated to stem from more than just the choice of optimizer. The proposed method enhances the diversity of sample learning by partitioning samples into different task combinations, while inner and outer loops facilitate the model in learning more varied feature patterns to improve overall generalization ability. The meta-learning method is compared with the original SemiCD method combined with various mainstream optimizers. The MLSCD method is compared with various optimization methods, as detailed in Table III. On the one hand, the meta-learning dual-optimizer method is compared with the widely used optimizer SGD and AdamW. The evaluation metrics clearly demonstrate the superiority of the MLSCD method, indicating that mainstream optimizers lacking a meta-learning mechanism fail to effectively learn perturbation task patterns and enhance model generalization. On the other hand, the method is compared with currently popular optimizers that transition with learning states, such as AMSGrad, SWATS, and HN-Adam. The experimental results further demonstrate that in semi-supervised change detection, the optimization advantage of the MLSCD method is not solely dependent on the choice of optimizer. The results are consistent with theoretical analysis. The meta-learning dual-optimizer method enhances the diversity of sample learning by partitioning sample combinations into diverse tasks. The inner and outer loops facilitate the model's ability to learn more varied feature patterns to significantly improve the model's

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TABLE I

DESCRIPTION OF COMPARISON METHODS FOR STATE-OF-THE-ART OPTIMIZERS

Method	Description
SGD [26], [27]	SGD is a popular method that updates model parameters using random samples to accelerate model convergence.
AWadam [30]	AWadam introduces a term to correct the variance of the adaptive learning rate, thereby reducing the variance in the early stages of training and improving stability and generalization performance.
Radam [61]	Radam introduces a term to correct the variance of the adaptive learning rate, thereby reducing the variance in the early stages of training and improving stability and generalization performance.
AMSGrad [62]	The AMSGrad method involves switching to SGD during the training process when the performance of Adam plateaus, in order to improve the model's generalization ability.
HN-Adam [63]	HN-Adam introduces optimized adaptive learning rate adjustment, taking into account the number of iterations and parameter update formulas, to improve training effectiveness.

TABLE II

DESCRIPTION OF COMPARISON METHODS FOR STATE-OF-THE-ART SEMI-SUPERVISED CHANGE DETECTION

Method	Description
s4GNA [8]	s4GNA combines high-level and low-level consistency to improve the performance of semi-supervised learning.
AdvNet [9]	AdvNet uses adversarial entropy minimization to reduce the distribution difference between the source domain and the target domain, thus improving the performance of semi-supervised learning.
SemiCD [10]	SemiCD utilizes consistency regularization to effectively leverage information from unlabeled multi-temporal remote sensing images to improve change detection performance.
RCL [12]	RCL employs reliable contrastive learning methods to address the challenge of reducing the dependence of semi-supervised models on labeled data.
RWS [11]	RWS combines the methods of strong consistency and weak consistency to improve the performance of semi-supervised learning.
C2F-SemiCD [13]	C2F-SemiCD is a coarse-to-fine semi-supervised change detection method based on consistency regularization.
DiffMatch [51]	DiffMatch is a semi-supervised change detection method guided by a vision-language model.
Ours	Ours is a semi-supervised method that applies meta-learning with dual optimizers to the semi-supervised SemiCD approach.

TABLE III

MLSSCD IS COMPARED WITH DIFFERENT OPTIMIZATION METHODS

Method	**IoU^c**	**F1**	**Kappa**	**Rec.**
LEVIR(5% labeled) → LEVIR				
Ours	**74.25%**	**84.63%**	**0.8411**	**83.43%**
AdamW	73.05%	82.16%	0.8175	81.13%
SGD	73.17%	82.36%	0.8195	81.67%
AMSGrad	65.41%	79.31%	0.7904	78.45%
SWATS	60.18%	75.66%	0.7496	74.35%
HN Adam	59.66%	74.83%	0.7419	73.66%
WHU(5% labeled) → WHU				
Ours	**65.16%**	**80.14%**	**0.7953**	**79.06%**
AdamW	63.02%	76.55%	0.7837	76.95%
SGD	62.70%	76.08%	0.7523	74.78%
AMSGrad	60.31%	74.38%	0.7364	73.07%
SWATS	58.72%	72.97%	0.7248	71.96%
HN Adam	55.34%	70.36 %	0.6955	69.19%

TABLE IV

MLSSCD IS COMPARED WITH DIFFERENT SEMI-SUPERVISED METHODS

Method	**IoU^c**	**F1**	**Kappa**	**Rec.**
LEVIR(5% labeled) → LEVIR				
Ours	**74.25%**	**84.63%**	**0.8411**	**83.43%**
DiffMatch	74.09%	83.92%	0.8351	82.95%
C2F-SemiCD	72.67%	81.80%	0.8124	80.55%
RWS	73.13%	82.42%	0.8166	81.17%
RCL	71.55%	81.05%	0.8067	80.20%
SemiCD	73.04%	82.36%	0.81.40	80.87%
s4GNA	65.41%	78.84%	0.7819	77.96%
AdvNet	57.12%	73.23%	0.7282	72.25%
WHU(5% labeled) → WHU				
Ours	**65.16%**	**80.14%**	**0.7953**	**79.06%**
DiffMatch	63.79%	79.37%	0.7901	78.74%
C2F-SemiCD	62.92%	77.17%	0.7691	76.24%
RWS	63.70%	78.67%	0.7804	77.30%
RCL	62.82%	75.36%	0.7483	74.12%
SemiCD	62.70%	76.08%	0.7523	74.78%
s4GNA	56.29%	70.32%	0.6958	69.01%
AdvNet	42.61%	65.36%	0.6470	64.25%

overall generalization performance. This demonstrates that the optimization advantage of the MLSCD method is not solely dependent on the choice of the optimizer, highlighting the efficiency of the meta-learning inner and outer loop mechanisms in utilizing samples effectively. Next, we will continue to verify whether it also has advantages compared to other semi-supervised methods.

Thereafter, the MLSSCD method is compared against state-of-the-art semi-supervised methods, including those based on the pseudo-label generation method as well as consistency regularization methods that do not utilize pseudo-labels. Various semi-supervised change detection methods are detailed in Table IV. The evaluation data from the table clearly indicates that the MLSCD method, based on consistency regularization, outperforms methods s4GNA, AdvNet, and C2F-SemiCD, which rely on pseudo-label generation.Consistency regularization methods are more effective because they require the model to produce consistent predictions on both original and weakly perturbed samples, thereby overcoming the issue of inaccurate pseudo-label generation that can mislead model training. Consistency regularization methods are more effective because these methods do not rely on generating pseudo-labels, thereby avoiding the risk of incorrect pseudo-labels misleading the model. By requiring the model to produce consistent predictions for both the original samples and their weakly perturbed versions, these methods help overcome the issue of inaccurate pseudo-label generation. Additionally, a significant limitation of pseudo-labeling is the potential inaccuracy of the pseudo-labels, which can lead to incorrect guidance for the model and hinder its learning process. In comparison, the MLSCD method outperforms the semi, RCL, RWS, and DiffMatch methods by utilizing random perturbation

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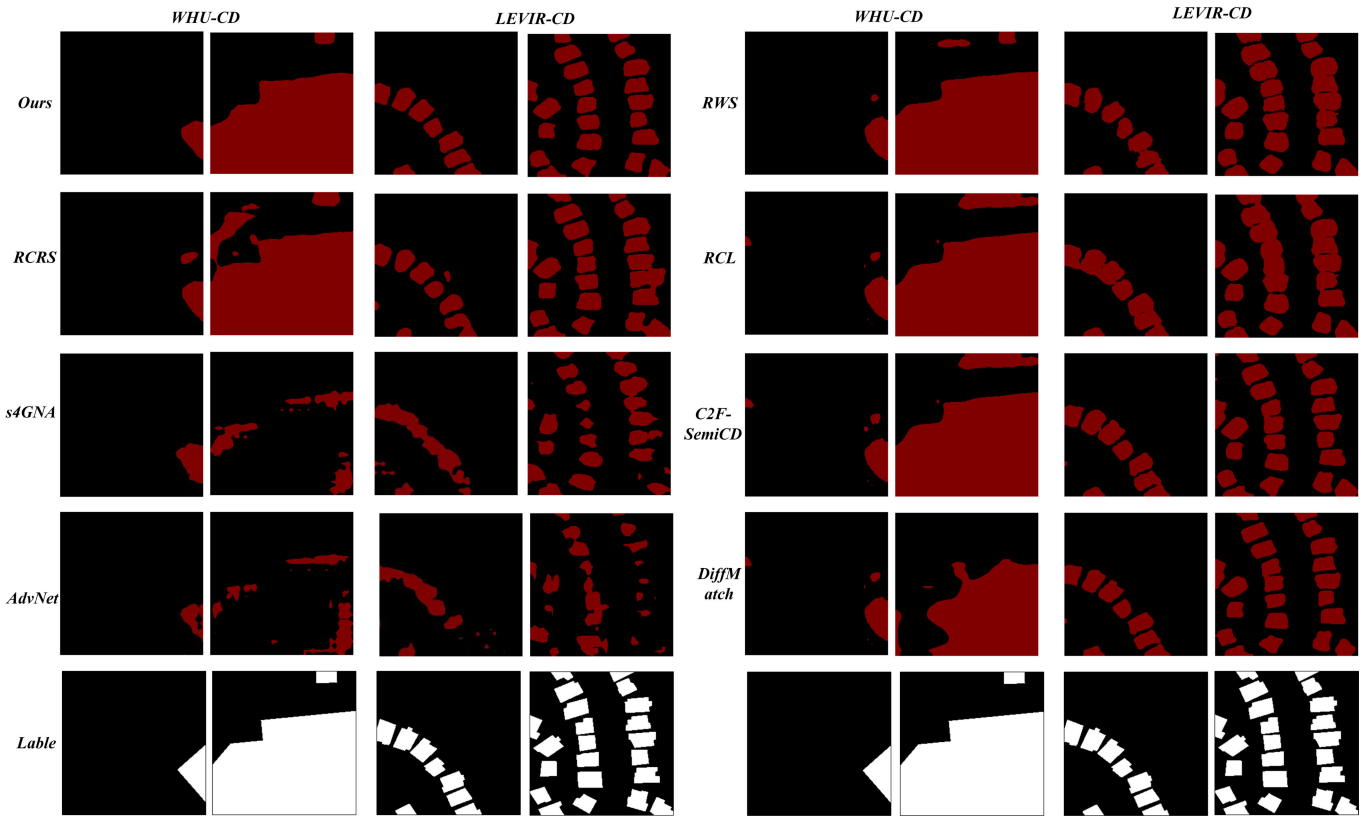


Fig. 4. Visualizations of the change detection results.

settings and multitask learning. The MLSCD method promotes diverse feature learning by partitioning individual samples into varied task-specific learning scenarios, enabling the model to better adapt to real-world variability and diverse perturbations. As a result, The proposed method enhances robustness and improves semi-supervised change detection performance.

Ultimately, the MLSSCD method is compared with current popular semi-supervised methods by visualizing the change detection prediction results. This analysis focused particularly on scenarios involving only 5% labeled samples. The detailed visualized results of change detection for different semi-supervised methods are shown in Fig. 4. The visualization results demonstrate that the MLSCD method achieves significantly higher detection accuracy. These experimental findings align with theoretical predictions, indicating that incorporating more realistic weak perturbations effectively increases the diversity of perturbation tasks. Moreover, dividing samples into multiple similar tasks improves learning efficiency. The meta-learning method enhances the model's stability and reduces the likelihood of false detections.

2) Ablation Analysis: The ablation study aims to validate the impact of labeled data quantity, optimizer choice, optimal ratio between the query set and the support set, and perturbation randomness on the MLSCD method. The following provides a detailed analysis of all influencing factors based on the experimental results.

The effect of the number of labeled samples on the MLSCD method is examined through experiments using labeled samples ranging from 5% to 40% of the total labels in the dataset. In this experiment, the SemiCD method and the Sup.only method are compared with the MLSCD method. The Sup.only method involves training exclusively with labeled samples. In contrast, the SemiCD method is trained using labeled samples and perturbed unlabeled samples, where the perturbation level is set to predefined values, without employing meta-learning methods. As shown in Table V, the MLSCD method achieves good performance with only 5%–40% of the labels, compared to the Sup.only and SemiCD methods. Comparing Sup.only and SemiCD demonstrates that incorporating unlabeled samples significantly mitigates overfitting and enhances change detection performance. Furthermore, a comparison with the semi-d method shows that partitioning into diverse tasks more effectively encourages the model to learn varied features, thereby enhancing generalization ability. Each perturbation of the unlabeled samples, combined with labeled samples, forms a learning task. Through meta-learning, the model learns the patterns of these perturbation tasks, enhancing its ability to handle unknown perturbations and thereby improving the model's generalization capability.

Later on, the impact of optimizer selection on meta-learning for semi-supervised change detection is examined, particularly focusing on a scenario with just 5% labeled samples. Four key variations are explored: Consistent regularization for semi-supervised change detection (SemiCD); SemiCD with meta-learning using SGD for both loops (SemiCD$_{(\mathrm{SGD,SGD})}$); SemiCD with meta-learning employing SGD for the inner loop and Awadm for the outer loop (SemiCD$_{(\mathrm{Awadm,SGD})}$); and SemiCD with meta-learning utilizing Awadm for both

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TABLE V

PERFORMANCE OF THE MLSCD METHOD IS VALIDATED THROUGH DIFFERENT PROPORTIONS OF LABELED SAMPLES

Method		IoU^c	F1	Kappa	Rec.
LEVIR(5%-40% labeled) → LEVIR					
Sup.only	5%	62.30%	77.23%	0.7657	74.89%
	10%	63.79%	78.01%	0.7746	75.80%
	20%	71.50%	80.35%	0.7754	76.33%
	40%	73.07%	84.27%	0.8292	81.47%
SemiCD	5%	73.47%	82.36%	0.8309	86.12%
	10%	73.06%	86.21%	0.8415	83.81%
	20%	75.02%	87.80%	0.8607	84.95%
	40%	75.63%	88.43%	0.8693	86.77%
Ours	5%	74.25%	84.63%	0.8411	83.43%
	10%	75.60%	87.40%	0.8661	86.47%
	20%	78.83%	88.64%	0.8809	87.54%
	40%	**80.22%**	**90.35%**	**0.8894**	**88.63%**
WHU(5%-40% labeled) → WHU					
Sup.only	5%	50.79%	73.53%	0.7253	72.53%
	10%	55.43%	76.95%	0.7650	75.11%
	20%	64.67%	80.19%	0.7954	79.46%
	40%	73.18%	83.93%	0.8327	82.93%
SemiCD	5%	62.70%	76.08%	0.7523	74.78%
	10%	64.27%	83.81%	0.8352	79.20%
	20%	67.04%	85.17%	0.8466	83.94%
	40%	70.49%	86.98%	0.8673	82.08%
Ours	5%	65.16%	80.14%	0.7953	79.06%
	10%	68.09%	85.81%	0.8461	84.29%
	20%	74.36%	87.54%	0.8681	86.37%
	40%	**79.26%**	**88.93%**	**0.8793**	**87.93%**

TABLE VI

EXAMINING THE IMPACT OF OPTIMIZER SELECTION ON META-LEARNING FOR SEMI-SUPERVISED CHANGE DETECTION

LEVIR(5% labeled) → LEVIR					
Method	IoU^c	F1	Kappa	Rec.	Time(h)
$(SemiCD_{(AwadmW,AwadmW)})$	**74.47%**	**85.21%**	**0.8474**	**84.03%**	35.19
$(SemiCD_{(SGD,SGD)})$	73.53%	81.16%	0.8071	79.87%	28.35
$(SemiCD_{(Awadm,SGD)})$	73.27%	82.64%	0.8205	81.77%	29.09
Ours	74.25%	84.63%	0.8411	83.43%	**29.40**
WHU(5%labeled) → WHU					
Method	IoU	F1	Kappa	Rec.	Time(h)
$(SemiCD_{(AwadmW,AwadmW)})$	**68.79%**	**84.11%**	**0.8373**	**82.50%**	28.20
$(SemiCD_{(SGD,SGD)})$	63.98%	81.45%	0.8107	80.89%	24.02
$(SemiCD_{(Awadm,SGD)})$	64.06%	82.74%	0.8212	81.93%	24.19
Ours	65.16%	83.53%	0.8314	82.76%	**24.58**

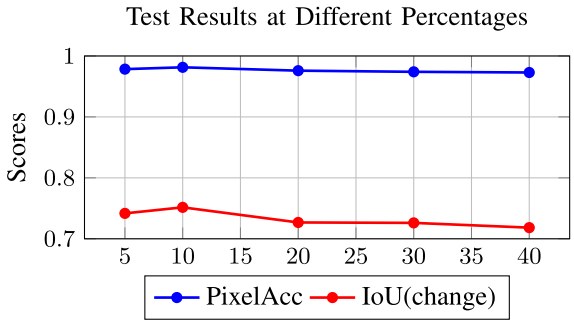


Fig. 5. Impact of different query set proportions on the results of meta-learning for change detection.

loops ($SemiCD_{(AwadmW,AwadmW)}$). The experimental results in Table VI reveal that the choice of the optimizer in meta-learning for semi-supervised change detection significantly impacts the learning of perturbation tasks. SGD is highly efficient due to its simplicity and low memory requirements. Conversely, AdamW, despite its higher computational complexity and increased memory consumption, often excels in complex optimization scenarios owing to its adaptive learning rates and effective weight decay mechanism. Unlike Adam, AdamW decouples weight decay from gradient updates, resulting in more robust regularization capabilities. In the MLSCD method, balancing rapid learning with performance optimization is crucial. Utilizing SGD in the inner loop facilitates swift adaptation to task-specific patterns, while AdamW in the outer loop effectively fine-tunes meta-level knowledge to maximize generalization. This dual-optimizer strategy takes advantage of the complementary strengths of SGD and AdamW, resulting in a more efficient and comprehensive optimization process.

After the appropriate optimizer is selected, the optimal ratio between the query set and the support set in meta-learning for change detection is validated. In this experiment, 5% of the data is labeled, while the rest is unlabeled for validation. Additionally, in the 5% labeled data, 5%, 10%, 20%, 30%, and 40% are selected as the query set to evaluate the effect of varying query set proportions on the MLSCD method. This MLSCD method allowed the optimal ratio between the query set and the support set to be evaluated for this specific task. Fig. 5 shows that the optimal performance is achieved when the query set constitutes 10% of the labeled data. In meta-learning, the model updates its parameters using the query set to enhance generalization, while the support set enables learning task-specific representations. However, if the query set is too large, the reduction in the support set can hinder the model's ability to effectively learn task-specific features, thereby reducing overall performance.

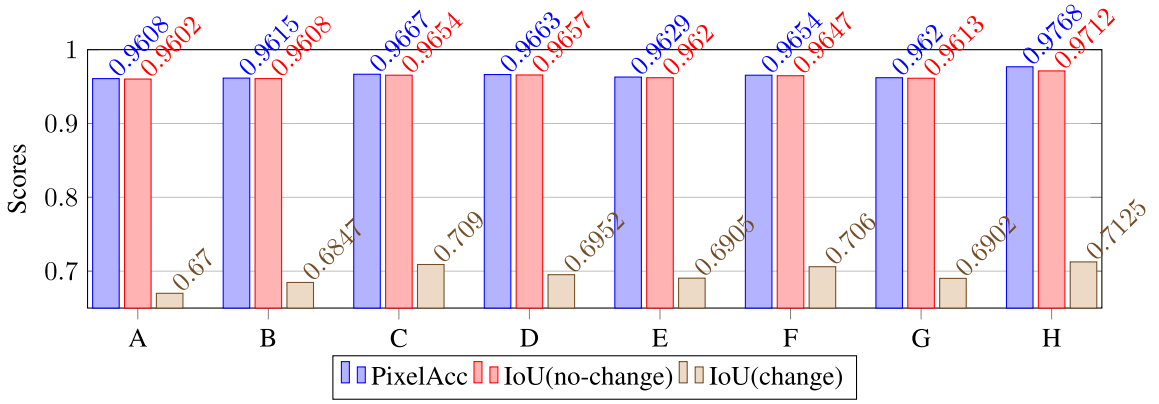
Eventually, the performance of the MLSCD method is further validated by investigating the effect of perturbation randomness on the change detection outcomes. All perturbation types listed in Table VII are experimentally evaluated to investigate the effects of different types of perturbations and their random combinations on detection performance. PixelAcc, IoU(no-change), and IoU(change) are employed as metrics to provide a comprehensive assessment of model performance, capturing overall pixel accuracy, the ability to detect unchanged areas, and effectiveness in identifying changes. As shown in Fig. 6, the impact of each type of perturbation on change detection is displayed. It also shows that relying on a single perturbation limits the model's ability to learn from diverse perturbations, which restricts the improvement of its generalization ability. In contrast, randomly selecting one perturbation from seven available types introduces greater task diversity, leading to significantly improved detection performance. Fig. 7 shows the experimental comparison results between nonrandom perturbations and random combinations, indicating that the latter achieves higher accuracy in change detection. The experimental results are consistent with theoretical expectations. In real-world scenarios, remote sensing images are typically affected by perturbations of varying intensity and type. Therefore, randomizing both perturbation strength and type better simulates real-world conditions, enhancing task diversity and significantly improving the model's generalization capacity.

3) Performance Evaluation: Validating the effectiveness of the proposed MLSCD method requires consideration of

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TABLE VII
DESCRIPTION OF DIFFERENT TYPES OF PERTURBATIONS

Perturbation Name	Operation Description	Mechanism
VATDecoder (Virtual Adversarial Training)	Generate adversarial perturbations to make the model more robust to unlabelled data.	Enhances robustness by finding the most sensitive perturbation direction and maximizing KL divergence.
DropOutDecoder (Dropout)	Randomly drop neurons or parts of feature maps to prevent overfitting and enhance generalization.	Prevents overfitting by forcing independent neural regions to predict without others' help.
CutOutDecoder (CutOut)	Randomly erase parts of the input feature map to improve robustness to missing information.	Improves model's capability to deal with partial information loss through data augmentation.
ContextMaskingDecoder (Context Masking)	Mask context regions in the input to force the model to rely on local information.	Forces reliance on remaining input, enhancing the model's ability to handle missing context.
ObjectMaskingDecoder (Object Masking)	Mask key objects or regions to force the model to rely on other information.	Improves performance when significant objects or information are absent.
FeatureDropDecoder (Feature Dropout)	Randomly drop features to reduce the model's dependence on specific features.	Enhances generalization by avoiding over-reliance on specific feature subsets.
FeatureNoiseDecoder (Feature Noise)	Add random noise to feature maps to improve the model's robustness to input uncertainty.	Improves stability and generalization by making the model handle noisy inputs.



A: vat decoder, B: drop decoder, C: cut decoder, D: object masking, E: feature noise, F: context decoder

F: context decoder, G: feature drop, H: randomly select one

Fig. 6. Effect of different perturbation types on meta-learning for change detection.

TABLE VIII
COMPARISON OF RUNNING TIME

WHU(unlabeled)LEVIR(5% labeled) → LEVIR								
Method	Ours	RWS	RCL	SemiCD	AdvNet	s4GNA	C2F-SemiCD	DiffMatch
Times(h)	29.40	29.09	28.42	27.25	30.34	30.07	24.57	25.33
LEVIR(unlabeled)WHU(5% labeled) → WHU								
Method	Ours(SemiCD)	RWS	RCL	SemiCD	AdvNet	s4GNA	C2F-SemiCD	DiffMatch
Times(h)	24.58	24.21	23.58	22.12	25.17	24.29	19.24	22.19

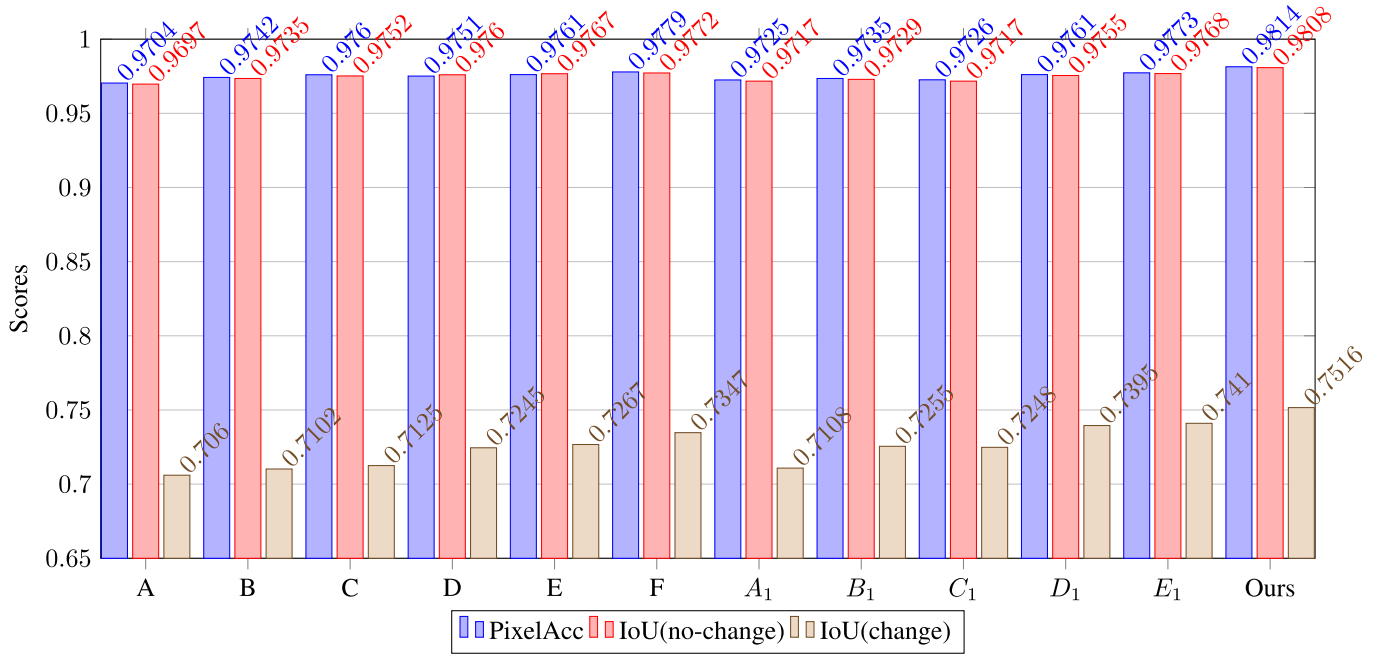
TABLE IX
TEST DATASET IS SUBJECTED TO THE INTRODUCTION OF NOISE, AND MODEL'S CAPACITY TO WITHSTAND NOISE IS ASSESSED

Method	**IoU^c**	**F1**	**Kappa**	**Rec.**
LEVIR(5% labeled) → LEVIR				
Ours	70.12%	78.68%	0.7825	77.81%
RCL	64.27%	74.42%	0.7380	73.26%
SemiCD	68.84%	76.56%	0.7614	75.70%
WHU(5% labeled) → WHU				
Ours	59.89%	74.26%	0.7385	73.27%
RCL	55.97%	66.77%	0.6612	65.79%
SemiCD	57.37%	69.71%	0.6939	68.91%

both change detection accuracy and training time. Table VIII presents the detailed training times for all methods, each evaluated over 80 epochs. Although incorporating the meta-learning method increases the training time, the overall impact remains minimal. All methods are run for 80 epochs, with total run times ranging from 23 to 24 h. Notably, the meta-learning method required approximately 30 additional minutes of runtime. Additional time is expected because the meta-learning method, which divides the learning into multiple diverse tasks to improve sample utilization efficiency and inevitably incurs some time costs. This method uses both inner and outer loop learning to understand perturbation patterns, which requires more training time. Overall, the minimal difference in runtime underscores the effectiveness of the meta-learning method in semi-supervised change detection.

Later, to evaluate the robustness of the proposed method against noise, Gaussian noise is added to the test samples. Noise with dimensions equivalent to the input tensor is generated using a normal distribution random number generator and then integrated into the original image tensor, with a standard deviation set at 0.3. The experimental results in Table IX indicate that the consistency regularization-based

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A: Two perturbations, B: Three perturbations, C: Four perturbations, D: Five perturbations, E: Six perturbations

F: Seven perturbations, A_1: Randomly combine two perturbations, B_1: Randomly combine three perturbations

C_1: Randomly combine four perturbations, D_1: Randomly combine five perturbations, E_1: Randomly combine six perturbations

Fig. 7. Impact of different perturbation combinations on meta-learning for change detection.

semi-supervised change detection method exhibits strong robustness to noise. These findings align with theoretical predictions. Meta-learning enhances sample utilization efficiency by dividing tasks into multiple similar subtasks, while perturbations increase the diversity of learning tasks, effectively improving the model's robustness. By learning the patterns of task diversity, the meta-learning method improves the stability of the model and reduces the likelihood of false detections.

V. Conclusion

Current consistency regularization methods lack randomness in perturbation settings and treat all samples as a single task, limiting the model's ability to learn from sample diversity and hindering improvements in its generalization ability. An MLSCD method for remote sensing images is proposed, addressing the limitations of traditional consistency regularization semi-supervised methods. The MLSCD method meta-learning enhances sample utilization efficiency by dividing tasks into multiple similar tasks, and random perturbation settings further increase the diversity of these learning tasks. A pivotal aspect of the MLSCD is the application of meta-learning to identify patterns within these varied tasks, thereby acquiring meta-knowledge. This meta-knowledge significantly enhances the model's ability to generalize to previously unseen real-world perturbations, thereby improving the overall performance of change detection. This method also confirms that balancing change detection accuracy and training time is crucial. It is optimal to use the AdamW algorithm for cross-task parameter updates in the meta-learning outer loop, along with the SGD algorithm for specific task optimization in the inner loop. The superiority of the proposed method is validated on two datasets. The extensive experimental results demonstrate the superior performance of the proposed method compared to state-of-the-art methods. Current semi-supervised change detection methods based on consistency regularization have demonstrated improvements in generalization capabilities. However, semi-supervised methods often rely heavily on large amounts of unlabeled data, leading to significant increases in runtime. Future research will focus on critical sample selection using support vector theory to explore how meta-learning change detection methods can reduce dependence on unlabeled data, decrease runtime, and improve overall efficiency.

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